

# Music Era and Release-Market Classification

DeepMIR 2026 Fall — Homework 1

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Code, checkpoints, README: <https://github.com/chijenpeng/26-fall-DeepMIR-HW1>

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## Experimental Setting

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## Pipeline

- 30 s clip, 24 kHz mono
- frozen encoder  $\rightarrow$  mean + std pooling over time
- StandardScaler  $\rightarrow$  L2 logistic regression
- train = fit, validation = model selection, test = prediction file only

## Hardware / software

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|           |                                                   |
|-----------|---------------------------------------------------|
| GPU       | 2 $\times$ NVIDIA RTX A4500 (20 GB)               |
| CPU / RAM | Intel i9-14900K, 125 GB                           |
| OS        | Ubuntu 26.04                                      |
| Software  | PyTorch 2.14, Transformers 4.45, scikit-learn 1.9 |

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## Results

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# Task 1: release decade

**Model:** frozen encoders + logistic regression

- Whisper-tiny encoder, **layer 1**
- MERT-v1-330M, **layers 5, 6, 14**
- mean + std pooling over time (6,912-d)

| Validation, 132 clips | Top-1        | Top-3        |
|-----------------------|--------------|--------------|
| Best model            | <b>52.3%</b> | <b>86.4%</b> |

## Errors

- **57%** of errors land on a **neighboring decade**
- Hardest: 2000s (recall .23), absorbed by 2010s and 1990s
- Easiest: 1960s and 2010s, which have one neighbor each

|      |       |           |          |           |          |          |           |
|------|-------|-----------|----------|-----------|----------|----------|-----------|
| True | 1960s | <b>17</b> | 2        | 0         | 3        | 0        | 0         |
|      | 1970s | 5         | <b>9</b> | 3         | 2        | 2        | 1         |
|      | 1980s | 2         | 2        | <b>13</b> | 2        | 3        | 0         |
|      | 1990s | 1         | 3        | 2         | <b>9</b> | 4        | 3         |
|      | 2000s | 2         | 2        | 1         | 5        | <b>5</b> | 7         |
|      | 2010s | 0         | 2        | 0         | 0        | 4        | <b>16</b> |
|      |       | 1960s     | 1970s    | 1980s     | 1990s    | 2000s    | 2010s     |
|      |       | Predicted |          |           |          |          |           |

Counts. Rows = true decade (22 clips each).

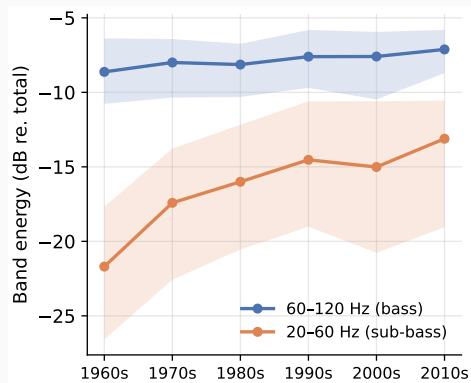
# Why neighbors get confused (1): the low end

**Hypothesis.** Old masters were cut for vinyl, where deep bass makes the stylus jump, so they should carry less sub-bass.

**Analysis.** Long-term average spectrum (Welch) of the 1,026 training clips, split into two bands.

| dB re. total | 60s   | 70s   | 80s   | 90s   | 00s   | 10s   |
|--------------|-------|-------|-------|-------|-------|-------|
| 20-60 Hz     | -21.7 | -17.4 | -16.0 | -14.5 | -15.0 | -13.1 |
| 60-120 Hz    | -8.6  | -8.0  | -8.1  | -7.6  | -7.6  | -7.1  |

- Sub-bass rises by 8.6 dB from the 1960s to the 2010s; the 60-120 Hz band barely moves
- From the 1990s on the low end is nearly identical, so it cannot separate 1990s / 2000s / 2010s



Median and inter-quartile range, training split.

## Why neighbors get confused (2): the loudness war

**Hypothesis.** From the 2000s, brick-wall limiters make masters louder and flatten peaks.

| Median per decade, training clips        | 60s   | 70s   | 80s   | 90s   | 00s   | 10s   |
|------------------------------------------|-------|-------|-------|-------|-------|-------|
| RMS level (dBFS)                         | -14.0 | -14.6 | -14.5 | -13.0 | -11.0 | -10.6 |
| Crest factor, peak – RMS (dB)            | 13.9  | 14.5  | 14.5  | 13.0  | 11.0  | 10.6  |
| Dynamic range, loudest 20% blocks (dB)   | 10.6  | 11.2  | 11.3  | 10.7  | 8.9   | 8.7   |
| Longest run pinned at the peak (samples) | 5     | 5     | 6     | 9     | 22    | 26    |
| Samples within 0.5 dB of the peak (%)    | 0.02  | 0.01  | 0.02  | 0.06  | 0.42  | 0.50  |

- 2000s and 2010s clips are 3 dB louder, have 3 dB less crest, and hold peaks at the ceiling. The two are **almost indistinguishable**, matching the 2000s ↔ 2010s confusion
- Production cues only split the data into “old” and “new”; finer distinctions must come from musical content

**Caveat.** The clips are 24 kHz mono, decoded from lossy Opus (YouTube). Resampling, downmixing and lossy coding reshape waveform peaks, so these numbers are trends (ANOVA  $p < 10^{-30}$ ), not exact properties of the original masters. The two sample-level rows are the most affected.

## Task 2: release market

**Model:** frozen encoders + logistic regression

- Whisper-large-v3 encoder, **layers 23, 30**
- MERT-v1-95M, **layer 7**
- Whisper-small language-ID posteriors

| Validation, 102 clips | Top-1        | Top-3        |
|-----------------------|--------------|--------------|
| Best model            | <b>60.8%</b> | <b>90.2%</b> |

### Errors

- **75%** of errors are among **US, UK, Germany, Italy**
- Largest confused pair: US ↔ UK (9 clips)
- Brazil (.88) and Spain (.82) are rarely missed
- Cause: English-language releases dominate Germany (83%) and Italy (55%)

| True | US      | 11        | 4  | 0      | 0     | 1       | 1     |
|------|---------|-----------|----|--------|-------|---------|-------|
|      | UK      | 5         | 9  | 0      | 0     | 2       | 1     |
|      | Brazil  | 0         | 0  | 15     | 0     | 0       | 2     |
|      | Spain   | 0         | 2  | 0      | 14    | 1       | 0     |
|      | Germany | 3         | 4  | 1      | 2     | 6       | 1     |
|      | Italy   | 1         | 4  | 0      | 2     | 3       | 7     |
|      |         | US        | UK | Brazil | Spain | Germany | Italy |
|      |         | Predicted |    |        |       |         |       |

Counts. Rows = true market (17 clips each).

## Optional Experiments

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Input length: 5 / 10 / 15 / 30 s

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# Ablation

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# Frozen features vs. fine-tuning

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# Audio Language Model

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# Disclosure

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## References

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